

Long-term evaluation of the pharmacokinetics of MEDI4893, an investigational anti-*Staphylococcus aureus* alpha toxin human monoclonal antibody, in healthy adults: a phase 1, randomized placebo-controlled study

X. Yu¹, G.J. Robbie², Y. Wu², M.T. Esser³, K. Jensen⁴, H.I. Schwartz⁵, M. Hernandez-Illas⁵, H.S. Jafri⁴

¹MedImmune, Gaithersburg, USA

²Clinical Pharmacology & DMPK, Gaithersburg, USA

³Translational Medicine, Gaithersburg, USA

⁴Clinical Development- MedImmune, Gaithersburg, USA

⁵Early Phase Clinical Trials- Miami Research Associates, Miami, USA

Objectives: MEDI4893 is an investigational immunoglobulin G monoclonal antibody that specifically binds to and neutralises alpha toxin (AT), a *Staphylococcus aureus* virulence factor. Anti-AT antibody prophylaxis has been associated with reduction in mortality in a *S. aureus* lethal pneumonia mouse model. A triple mutation M252Y/S254T/T256E (YTE) was engineered into the Fc region to extend the terminal half-life. The objective of this study was to evaluate the safety, tolerability, pharmacokinetics (PK), and anti-drug antibodies (ADAs) response profile of MEDI4893 following a single intravenous (IV) infusion in healthy adult volunteers.

Methods: This first-in-human phase 1 randomised, double-blind, placebo-controlled, single-dose, dose-escalation study was conducted at a single centre (NCT01769417). Males (n=17) and females (n=16) aged 22 to 57 years were enrolled into 1 of 4 dosing cohorts and randomly assigned to MEDI4893 (M) or placebo (PBO) as follows: cohort 1 (M 225 mg=3; PBO=1); cohort 2 (M 750 mg=3; PBO=1); cohort 3 (M 2250 mg=8; PBO=2); or cohort 4 (M 5000 mg= 12; PBO=3). Safety assessments included monitoring adverse events (AEs) and measuring laboratory parameters. Serum samples were obtained at prespecified time points for evaluation of PK, and ADA, up to 360 days after dosing.

Results: No serious AEs or AEs leading to discontinuation of study drug were observed during the study. MEDI4893 serum exposure increased dose-proportionally; the mean maximum observed concentration (C_{max}) increased from 77.2 µg/mL at 225 mg to 1784 µg/mL at 5000 mg, and the area under the concentration-time curve from time 0 to 360 days after dosing (AUC_{0-360}) increased from 4,914 µg*day/mL at 225 mg to 99,365 µg*day/mL at 5,000 mg. Serum clearance of MEDI4893 ranged from 42 to 50 mL/day. MEDI4893 terminal half-life ($t_{1/2}$) was estimated to be 80 to 112 days. A total of 3 adults tested positive for ADA before dosing and 3 adults postdosing only on day 151, but not beyond through 360 days postdose and with no effect on PK or safety.

Conclusions: Following a single IV dose, MEDI4893 exhibited linear and dose proportional PK across the dose range of 225 mg to 5000 mg. Serum half-life of 80-112 days is approximately 4-fold longer compared to other human IgG antibodies, confirming the extension of the terminal half-life by the YTE mutation. ADA rates were low and did not affect PK or safety. The safety data support continued development of MEDI4893.

Funding: This study was sponsored by MedImmune.